ISE 201 Math Foundations for Data Science and Decision Science

Introduction

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ISE @ SJSU
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Hi my name is Dr. Shilpa Gupta , and welcome to ISE 201 Math Foundations for Data Science and Decision Science

Data

Introductions

What domain have you worked in or familiar with?

What questions you were trying to answer with data?

Science of Data

Collection, analysis, interpretation and presentation of data to:

- Make decisions
- Solve problems
- Design (or improve) product and /or processes

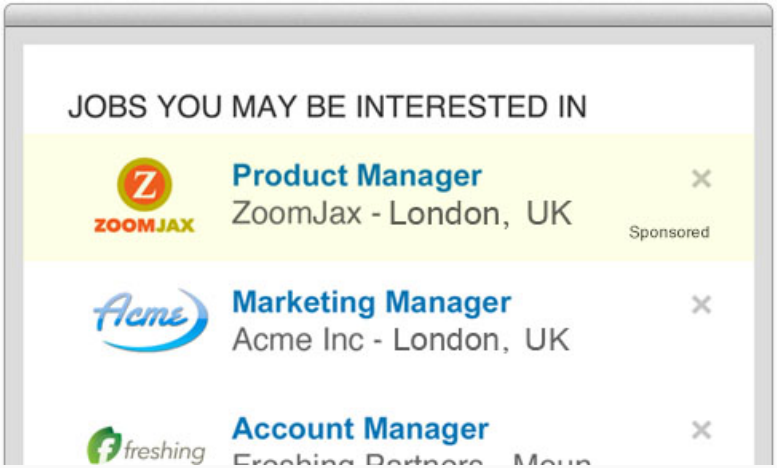
Data Science, Decision Science, ...

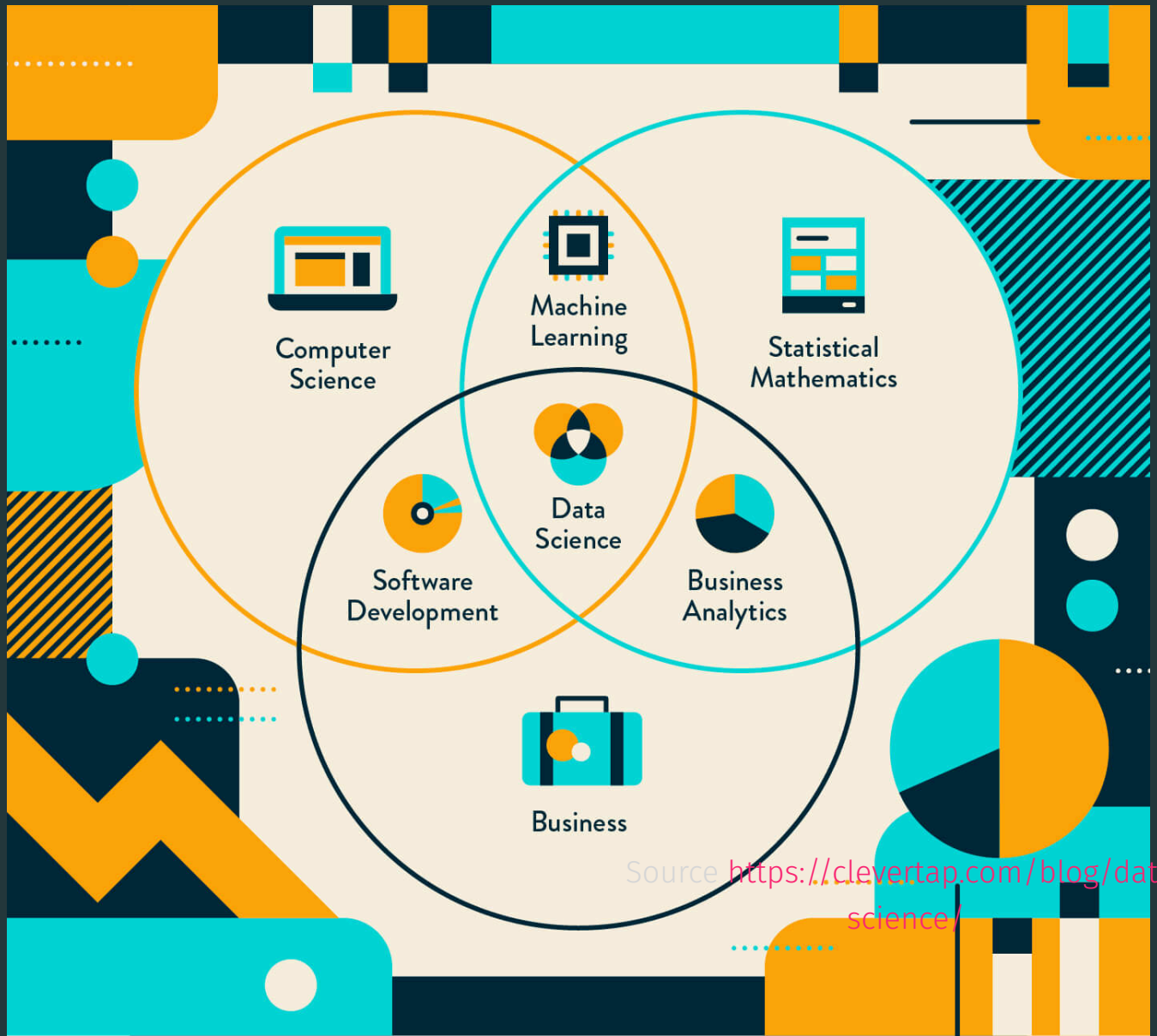
Business Analytics, Data Mining, Machine Learning, ...

What is the difference?

Key Difference

How the results of data {wrangling, summarization, prediction} are consumed and **by whom**





Source: https://clevertap.com/blog/data-science/

Let's talk about you

What aspect of data science you think you are good in?

What aspect of data science are you excited to learn?

Course Logistics

Purpose

- Descriptive Statistics and graphical methods to summarize and visualize data
- Linear Algebra concepts to discover variable-response relationships
- Statistical Inference to make decisions based on samples
- Probability concepts to quantify the risk in statistical inference
- Predictive models and optimization methods for finding the best approximation to the data

Course Logistics

Not Learning

1. Advanced models or Data Mining or NN or ML algorithms
2. Automating models
3. Deploying models
4. Agentic AI

Practice Thinking Skills

Divergent Thinking

Generate a wide range of ideas and think broadly and creatively

Convergent Thinking

Analyze and synthesize information and ideas while moving toward an answer or conclusion

Metacognition

Think about one's own thinking and learning

Metacognition

Self Reflection prompts

- How have I accomplished the reading/ task in the past?

- What do I already know about the material prior to reading?

- How has my prior understanding changed after reading?

- How have i learnt best in the past?

- What aspects of learning come easily to me and which ones are challenging?

- How can structure my day to acomplish my tasks?

...

Course Logistics

Email: shilpa.gupta@sjsu.edu

Office Hours: By appointment

Class: Thursdays 10:30am - 1:15pm

Book

1. Sign in or create an account at learn.zybooks.com
2. Enter zyBook code: SJSUISE201GuptaSpring2026
3. Subscribe

CANVAS

DO NOT email me submissions

Course Logistics

Assessments

- Python assignments as homework
- Quiz after end of each section
- Final Project
- Discussion Boards

Honor Code

NOT OK to ask someone to do assignments/projects for you.

NOT OK to copy solutions from classmates.

NOT OK to pretend that someone's solution is yours.

NOT OK to post your assignment solutions online.

OK to search, ask in public about the systems we're studying. **Cite all the resources you referenced.**

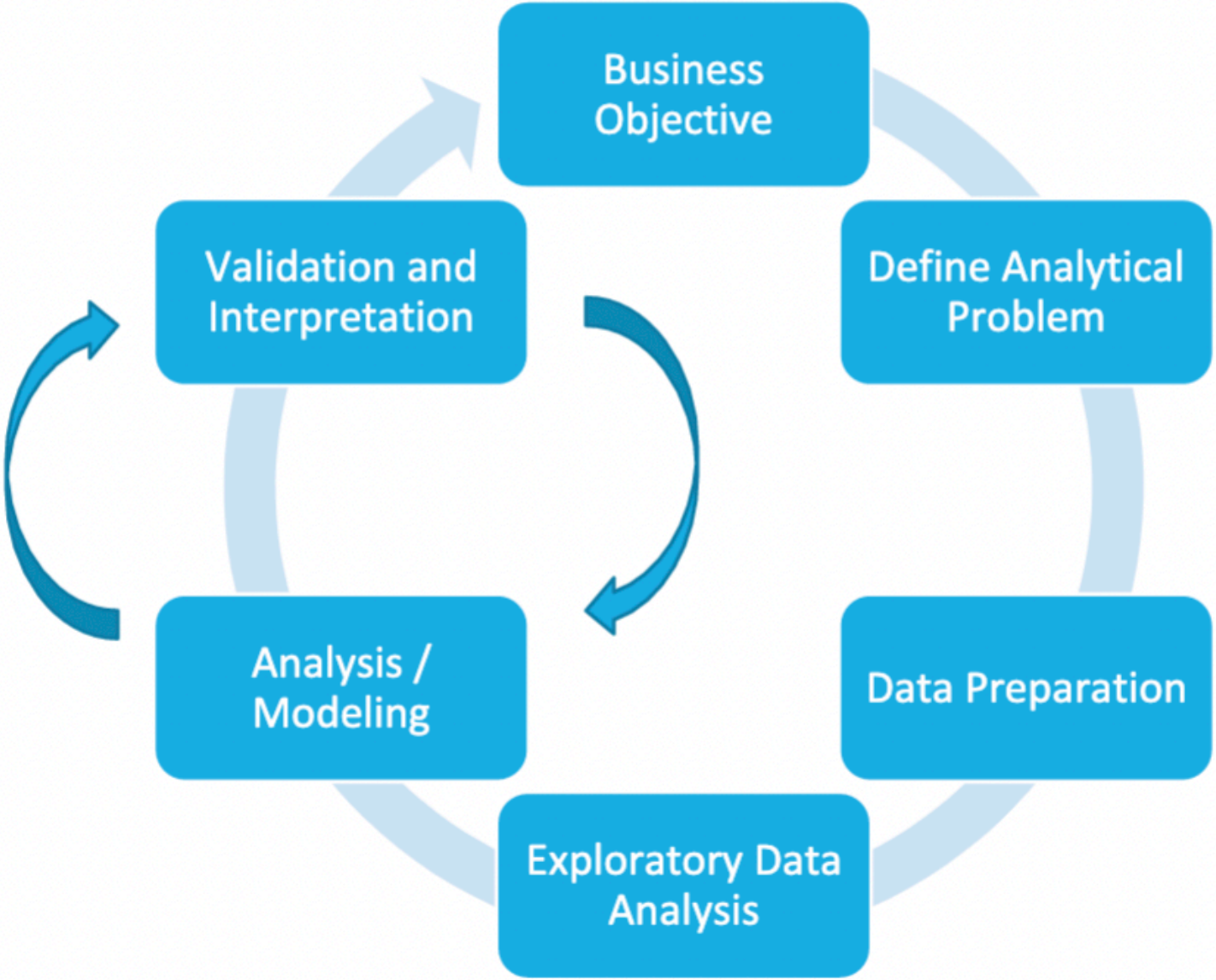
OK to discuss with classmates. **Disclose your discussion partners.**

ASK if unsure!

credit

Questions?

Data Science Process



Statistical Thinking

- Sources of variability
- Explaining or reducing variability

Population vs Sample

Population: Set of all items or events of interest

Sample: Subset of the population

Statistical Inference

Make valid conclusions about the population by observing the sample

A sample is **representative** and **random**

Free from over-representation or under-representation of a subgroup.

Every entity has equal chance to be selected.

Is this sample representative?

A university research team wants to estimate the average wheat yield per acre in a county. The county has 2,000 wheat farms, ranging in size from small family farms (10–50 acres) to large commercial farms (500+ acres).

The researchers collect data from 50 farms, chosen by asking for volunteers at the local farmer's market. Data collected: average yield from each farm, farm size, fertilizer usage

src: ChatGPT. OpenAI, 2023. https://chat.openai.com/chat

Is this sample biased?

A company is testing a new fitness tracking app. They want to know how satisfied users are and whether the app motivates people to exercise more.

They send a survey link only to premium subscribers (people paying for the upgraded version). Out of 10,000 total users, only 500 premium users respond. The results show 95% satisfaction and that most users exercise at least 5 times per week.

src: ChatGPT. OpenAI, 2023. https://chat.openai.com/chat

Collecting Data

- Retrospective study
- Observational Study
- Designed Experiment

Collecting Data

- Retrospective study

- Historical process data over same period of time
- Helpful in understanding correlation relationships
- Limited by amount and type of data collected

- Observational Study

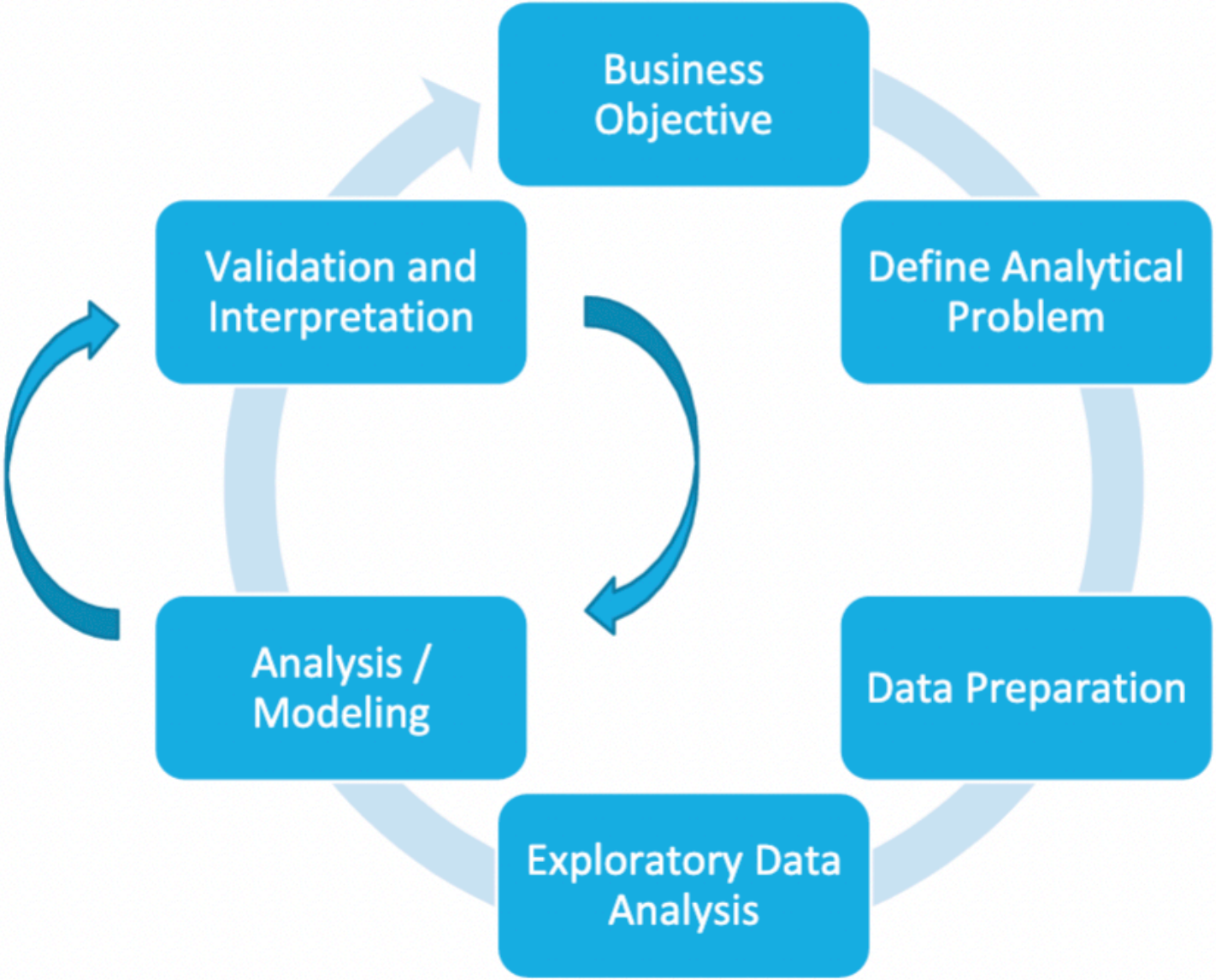
- Limited in how long the data is collected for
- Include additional quantities not measured in retrospective study

- Designed Experiment

- Deliberate and purposeful changes
- Helpful in understanding causal relationships

Questions?

Data Science Process



Asking Questions

"Far better an approximate answer to the **right question**, which is often vague, than an exact answer to the wrong question, which can always be made precise." - John Tukey

Question Formulation Technique*

1. Ask as many questions as you can
2. Do not judge or answer or discuss any of the questions yet
3. Change any statements into questions
4. Prioritize
5. Reflect and Repeat

*Rothstein, D., & Santana, L. (2011). Make just one change: Teach students to ask their own questions. Harvard Education Press.

Generating Data Questions

Descriptive

What is the summarized statistic of a dataset?

Exploratory

Are there any patterns, trends or relationships within a dataset?

Inferential

What can be inferred about the population from the sample?

Predictive

What would the outcome be for a certain combination of features ?

Causal

How does change in the levels of one factor causes changes in the other factor?

Example

Lets say we have farm dataset with values on crop type, acres planted, fertilizer used, rainfall, yield

Questions:

a. If a farmer increases fertilizer by 10 kg/acre, will this cause a significant increase in yield, holding rainfall constant?

b. Can we predict next season's crop yield for a farm given its acres, rainfall, and fertilizer inputs?

c. What is the average crop yield per acre across all farms in the dataset?

d. Based on our sample of farms, can we conclude (with 95% confidence) that fertilizer use increases average yield in the county?

e. Does yield vary systematically with rainfall and fertilizer use across different crop types?

Can we identify the type of questions?

src: ChatGPT. OpenAI, 2023. https://chat.openai.com/chat

Communicating Results

Goal: Communicate* information from data

1. Exploration phase for Self

2. Exploratory data analysis for generating Consensus

3. Storytelling with data to Communicate Insights with context

As we go from $1 \rightarrow 3$, the impact of the visualization is increasing and so is the number of people seeing it.

Communicating Results

Literate Programming

A computational document to include explanation in natural language and snippets of source code

So both *human* and *machine* can understand

Jupyter Notebook